4 (a) (i) State **one** difference and **one** similarity between the oscillations of a stationary sound wave and a progressive sound wave.

Difference: ● In stationary sound waves, particles oscillate (except at the nodes where it is 0) with varying amplitudes from 0 to a maximum at the antinodes. In progressive sound waves, all particles oscillate with the same amplitude.

● In stationary sound waves, particles between adjacent nodes oscillate in phase while particles in adjacent segments oscillate in antiphase. In progressive sound waves, particles oscillate at different phases within one wavelength [B1]

Similarity: In both waves, particles are oscillating in simple harmonic motion/about their equilibrium positions with the same frequency as the wave [B1]

[2]

(ii) Fig. 4.1 represents the displacement-time graph of two progressive sound waves travelling together through the same space.

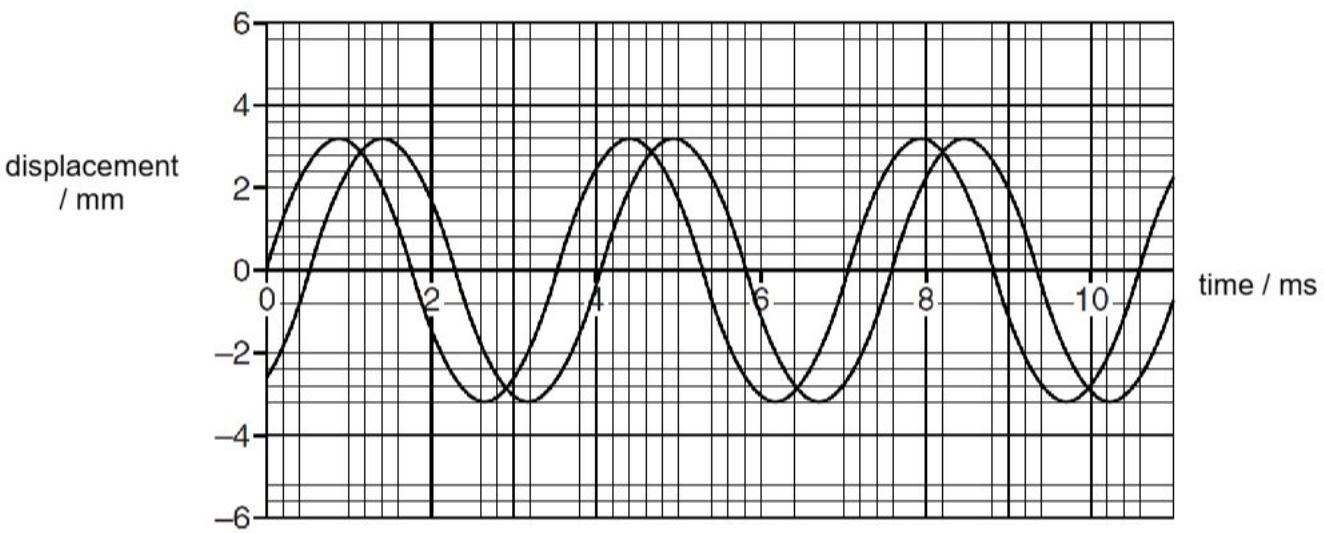


Fig. 4.1

1. State and explain whether Fig. 4.1 shows that the waves are coherent.

The waves are coherent because there is a constant time lag / constant period / constant frequency which indicates a constant phase difference [B1]

.....

.....

.....[1]

2. Use Fig. 4.1 to determine the phase difference between the waves.

$$\Delta\phi = \frac{\Delta t}{T} \times 2\pi$$

$$= \frac{0.5}{3.5} \times 2\pi \text{ OR } \frac{2.5 \text{ sq}}{17.5 \text{ sq}} \times 2\pi$$

$$= 0.898 \text{ rad [A1]}$$

phase difference = ... rad [1]

(iii) A sound wave travels along the axis of a tube that is open at both ends, as shown in Fig. 4.2. A stationary sound wave is formed in the tube.

The lowest frequency possible for the stationary sound wave in this tube is *f*.

On Fig. 4.2, mark all the positions of the displacement nodes (use the letter N) and the displacement antinodes (use the letter A) of a stationary sound wave for a frequency of 2*f*.

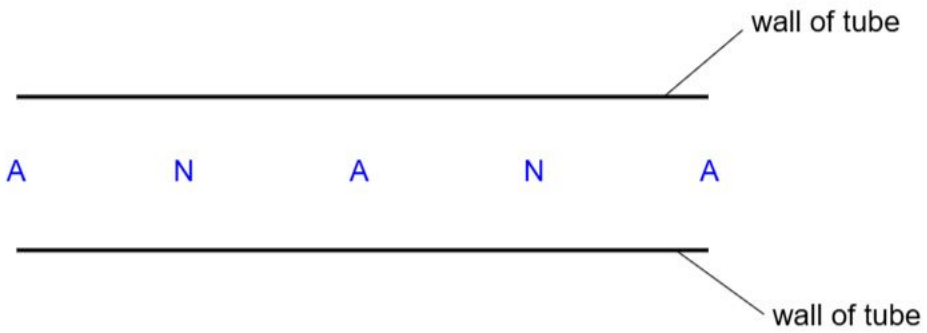


Fig. 4.2

[1]

(iv) With reference to the movement of air molecules on either side of a displacement node, explain why there is maximum variation in pressure at the nodes.

Particles on either side of a displacement node are moving in opposite directions [B1] so at one moment the pressure increases to a maximum (particles on either side are closest to node) AND at the next moment the pressure decreases to a minimum (particles on either side furthest from node) therefore pressure varies from a maximum to minimum → hence maximum variation [B1]

[2]

(b) Fig. 4.3 shows the arrangement for an experiment involving superposition of microwaves.

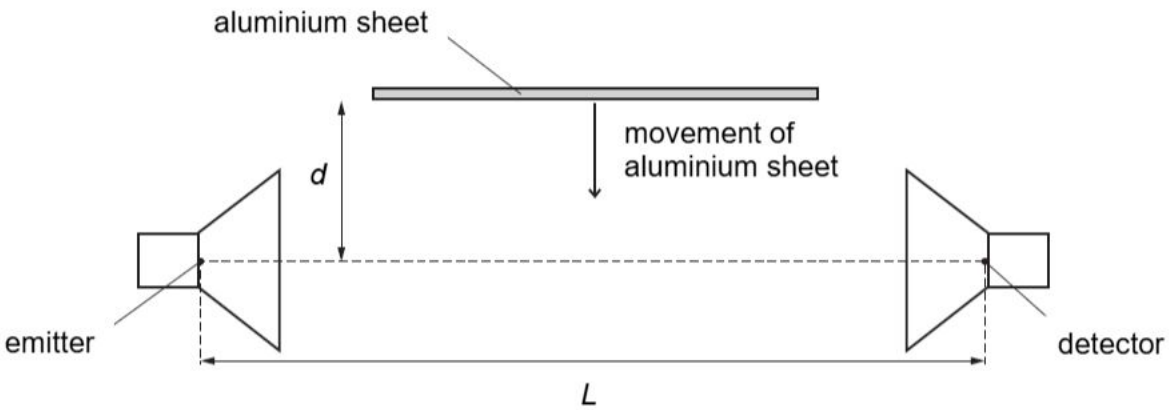


Fig. 4.3 (not to scale)

A microwave emitter and detector face each other with a distance *L* apart. A reflecting aluminium sheet is placed at a vertical distance *d* above the direct path between the emitter and the detector.

(i) When the aluminium sheet is moved in the direction shown by the arrow, the intensity of microwaves received by the detector fluctuates.

Describe and explain this variation in the microwave intensity.

Reducing d changes the path difference between the direct and reflected wave so the phase difference changes [B1]

Constructive interference leading to maximum intensity (occurs when path difference is a whole number of wavelengths) AND destructive interference leading to minimum intensity (occurs when path difference is odd number of half-wavelengths) [B1]

.....[2]

(ii) As d decreases, the microwave detector has a maximum reading at d = 21.2 cm and the next maximum at d = 17.7 cm. The distance L stays constant at 80.0 cm.

Determine the wavelength λ of the microwaves.

$$\text{reflected path} = 2\sqrt{\left(\frac{L}{2}\right)^2 + d^2}$$ [C1] Use of Pythagoras theorem to calculate reflected path

$$\text{path difference} = 2\sqrt{\left(\frac{L}{2}\right)^2 + d^2} - L$$

$$2\sqrt{\left(\frac{0.800}{2}\right)^2 + 0.212^2} - 0.800 = (n+1)\lambda \quad \text{---(1)}$$

$$0.1054 = (n+1)\lambda$$

$$2\sqrt{\left(\frac{0.800}{2}\right)^2 + 0.177^2} - 0.800 = n\lambda \quad \text{---(2)}$$

$$0.0748 = n\lambda$$

[M1] for correct path differences seen

$$\lambda = 0.1054 - 0.0748 = 0.0306 \text{ m}$$ [A1] correct method leading to correct ans

λ = ... m [3]

(iii) Microwaves undergo a phase change of π rad when they reflect.

When d is much less than the wavelength of the microwaves, it is observed that the detector reading is a minimum.

Suggest why this observation demonstrates that microwaves must undergo a phase change on reflection from the aluminium sheet.

When d is much less, path difference becomes negligible or almost zero (as distance travelled by reflected wave is almost similar to incident wave) [B1]

This is supposed to produce a maximum reading due to constructive interference, but since a minimum was detected, it implies a phase change of π rad had occurred due to reflection [B1]

.....[2]

[Total: 14]